Reflection on a cardiology rotation: a patient with chest pain

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1. Description

On Tuesday evening 4 November I was on the evening shift with the cardiology resident. At 22:35 the nursing staff called about Mr De Groot (62, name changed), who had been admitted that afternoon for assessment of cardiac arrhythmias. For the past fifteen minutes he had had pressing chest pain radiating to his left arm. The resident was busy in the emergency department and asked me to go and see the patient first.

I started by taking a thorough history: nature and duration of the pain, radiation, previous complaints, risk factors (smoking, hypertension, family history), medication and allergies. I then carried out a physical examination, including blood pressure in both arms to rule out aortic dissection (152/94 left, 148/92 right). After about twenty minutes I had an ECG made. It showed ST elevation in II, III and aVF. I called the resident, who came immediately and activated the cath lab. Mr De Groot turned out to have an inferior myocardial infarction and underwent successful angioplasty.

2. Feelings

I felt proud that the resident trusted me to see the patient on my own. During the history I felt calm and confident. When I saw the ECG I was startled, and briefly unsure whether I had interpreted it correctly.

3. Evaluation

What went well was my thoroughness. I took a complete history and did not miss any differential diagnosis: acute coronary syndrome, pulmonary embolism, aortic dissection, pneumothorax and pericarditis were all considered. I actively measured the blood pressure difference between the arms. Afterwards the resident said my handover was “very complete”.

What went less well was my confidence in assessing the ECG.

4. Analysis

According to the literature, a systematic approach to chest pain is essential to avoid missing life-threatening causes. That is what I did. My uncertainty about the ECG comes from having seen few ST-elevation infarctions during my rotations so far. I should have interpreted the ECG straight away instead of calling the resident.

5. Conclusion

In this situation I showed that I can assess a patient with chest pain thoroughly and systematically. The only point for improvement is my confidence in reading ECGs.

6. Action plan

Over the coming weeks I will practise five ECGs a day with an online module. Next time, with a clear ST-elevation infarction, I want to make the diagnosis myself and activate the cath lab, so that no time is lost calling the resident.